

SNAKE GAME PROJECT

Week 2 – Software Design

Project Name: SnakeGame
Java Swing | NetBeans | GitHub

1. Introduction

Week 2 focuses on the Software Design of the SnakeGame project. After completing the planning and requirement analysis in Week 1, the next step is to decide how the application will be structured. In this stage, the basic structure of the SnakeGame application, the main classes or components, their responsibilities, and the relationship between the user interface and game logic are planned. The basic workflow of the game is also prepared.

2. What I Need to Do

- **Design the overall structure of the SnakeGame application:** The first task of Week 2 is to design the overall structure of the SnakeGame application. This means deciding how the different parts of the application will be organized and how they will work together. A clear structure will make the project easier to develop and manage in the later weeks.
- **Plan the main classes/components and their responsibilities:** The main classes or components of the application need to be planned. Each class or component should have a clear responsibility within the project. This planning helps keep the software structure organized and makes it easier to understand which part of the application will handle a particular task.
- **Plan the relationship between the user interface and game logic:** The relationship between the user interface and the game logic needs to be planned. The user interface will provide the part through which the user interacts with the application, while the game logic will handle the main game operations. Planning their relationship helps ensure that the interface and the game functions can work together properly.
- **Prepare the basic workflow of the game:** A basic workflow of the SnakeGame needs to be prepared. The workflow describes the general sequence of how the game will operate. It provides a simple plan for how the application will move from one stage or operation to another.
- **Decide how the board, snake, food and controls will be represented:** The design should also decide how the main game elements will be represented. This includes the game board, the snake, the food and the controls. Planning these representations before implementation provides a clear structure for developing the actual game later.

3. Basic Software Structure

The Week 2 design creates the basic structure that will be used for the SnakeGame application. The application is planned around the main game elements and their responsibilities. The design includes the user interface, game workflow, board, snake, food and controls. These elements are planned at the design stage so that they can be implemented step by step in the following weeks.

4. User Interface and Game Logic Relationship

The user interface and game logic are two important parts of the application. The interface provides the interaction area for the user, while the game logic is responsible for the actual operation of the game. Week 2 focuses on planning how these two parts will communicate with each other. This design will later help connect user actions and game operations in a structured way.

5. Basic Game Workflow

1. Start the SnakeGame application.
2. Prepare the main game structure.
3. Represent the game board.
4. Represent the snake and food.
5. Provide the required controls.
6. Connect the interface with the game logic.
7. Continue the game according to the planned workflow.

The above workflow represents the basic design idea for the project. It is a planning stage, so the main purpose is to define how the different parts of the game will be organized before the actual implementation is developed.

6. What I Need to Learn

- **Basic Java class design:** Learn how Java classes can be designed and organized so that different parts of the application can be represented clearly.
- **Java Swing structure:** Learn the basic structure of Java Swing because the SnakeGame application will use Java Swing for its graphical interface.
- **Event-driven programming:** Learn the basic idea of event-driven programming so that user actions and application events can later be handled properly.
- **Basic project architecture:** Learn how to organize the different parts of a software project and understand how the main components can work together.

7. GitHub Upload / Progress

- **Software Design Document:** The software design information prepared during Week 2.
- **Basic class/module structure:** The planned basic structure of the classes or modules used in the project.
- **Updated README:** The README file updated with the Week 2 project progress.

8. Expected Week Result

At the end of Week 2, the basic software structure and workflow of the SnakeGame project are planned. The main classes or components, their responsibilities, the relationship between the user interface and game logic, and the representation of the board, snake, food and controls are considered as part of the software design. The required design materials can then be maintained in GitHub.

9. Week 2 Summary

Week 2 is the Software Design stage of the SnakeGame project. The main purpose of this week is to plan the structure of the application before implementation. The overall structure, main classes or components, user interface and game logic relationship, basic workflow, and representation of the board, snake, food and controls are planned. The main learning topics are basic Java class design, Java Swing structure, event-driven programming and basic project architecture. The Software Design Document, basic class/module structure and updated README are prepared as the Week 2 GitHub progress.